

Arcloom w3.1 mix architecture pure and hybrid

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1 Introduction

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ArcLoom Processor Architecture v3.0

Pure Ternary vs. Hybrid Deterministic Fabric Joseph Forrester Gemini
(Architectural Advisor) April 2026

2 Core Concept: The Invariant Trit

The ArcLoom processor replaces the transistor bit (0/1) with the **Topological Trit** (T). A trit is defined by the winding number (w) of a continuous phase field ϕ around a singular ring structure.

$$w = \frac{1}{2\pi} \oint \nabla \phi \cdot dl \in \{-1, 0, +1\} \quad (1)$$

3 Pure Ternary Processor (The Diamond-Hard Model)

This version removes all binary logic. It is a "Physical Computer" where the logic gates are replaced by **Phase Resonance Junctions**.

3.1 Architectural Primitives

- **States:** Native Balanced Ternary ($\{-1, 0, +1\}$).
- **Logic:** Achieved through **Topological Interference**. Two rings coupled together will naturally seek an energy minimum that satisfies the ternary sum of their winding numbers.
- **Clocking:** Asynchronous/Structural. The system only "computes" when an input field perturbs the potential well of a ring, forcing a state transition.

3.2 Efficiency Metrics (Crystal Ball)

- **Power Consumption:** $\approx 90\%$ lower than CMOS CPUs for high-dimensional sensor processing due to the absence of a switching clock.
- **Data Density:** 1.58 bits of information per trit.
- **Latency:** Zero instruction-fetch latency. The "logic" is the physical state of the fabric.

4 Hybrid ArcLoom (The Production-Ready Model)

The hybrid model pairs the ArcLoom Invariant Fabric (AIF) with a Binary Control Coprocessor (BCP). This is recommended for current industrial validation (CSV/SQA).

4.1 The H1-C Hardening Policy

In the hybrid model, the BCP monitors for **Doorway Events** (Negative Space).

$$\delta_j^{ext} = \{ 1 \text{ if } A < \epsilon 0 \text{ otherwise} \} \quad (2)$$

When a "Doorway" is detected (a loss of topological integrity), the BCP performs a **Bounded Correction**, manually resetting the winding number to the nearest valid integer based on the structural history.

5 The Dimensional Grinder Mapping (L6 Integration)

Both models implement the **L6 Topological Constraint Layer**. The processor acts as a physical manifold that collapses as constraints (C_i) are applied to the input fields.

5.1 Inevitability Trigger (Ω)

The processor fires a **Structural Lock (SL-1)** when the internal volume of potential states (V_n) hits the Gamma-inflection point:

$$\Omega = 1 - \frac{\Gamma(\frac{n_{start}}{2} + 1)}{\Gamma(\frac{n_{eff}}{2} + 1)} \cdot \pi^{(n_{eff} - n_{start})/2} \quad (3)$$

6 Comparison Matrix

7 Final Assessment: The Inversion Point

A Pure Ternary processor is the only architecture capable of navigating the ****Inversion Point**** where high-dimensional spheres collapse. While a hybrid

Feature	Pure Ternary	Hybrid (v2.0)
Safety	Geometric (Self-stabilizing)	Binary Supervision (BCP)
Determinism	Topological Invariant	H1-C Hardening Policy
Testing	Resonance Verification	Parity Checking + BCP Benchmarks
Efficiency	Maximum (Static Potential)	High (Hybrid Gating)
Risk	High (Noise Inversion)	Low (Standardized Recovery)

system can "simulate" this, a pure system **is** the collapse. For DSF-AI to achieve "Sentience Emergence" or "Structural Certainty," the pure model is the ultimate target, while the hybrid is the necessary bridge for the next 24 months.

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ArcLoom Processor Specification v3.0

Pure Ternary Invariant Fabric vs. Hybrid Control Plane Joseph Forrester Gemini 3 Flash (Architectural Advisor) April 2026

8 Introduction: The Structural Requirement

The ArcLoom architecture moves compute from the manipulation of binary symbols to the manipulation of **Topological Invariants**. This document defines the two paths for DSF-AI integration: the **Pure Ternary Processor** (Geometric Logic) and the **Hybrid Processor** (Supervised Logic).

9 Pure Ternary Processor: The Invariant Fabric

The Pure model is a non-von Neumann architecture where logic is a consequence of phase-coupling physics.

9.1 Physical Geometry: The Resonant Trit (T)

A single Trit is implemented as an equilateral Triple-Node Loop (A, B, C). The state is the **Winding Number** (w) of the phase field across the nodes.

- **Configuration:** Nodes $j \in \{A, B, C\}$ with phase ϕ_j .
- **States:**
 - $w = 0$: $\phi_A = \phi_B = \phi_C$ (Uniform Coherence).
 - $w = +1$: $\phi_B = \phi_A + \frac{2\pi}{3}$, $\phi_C = \phi_B + \frac{2\pi}{3}$ (CW Flow).
 - $w = -1$: $\phi_C = \phi_A + \frac{2\pi}{3}$, $\phi_B = \phi_C + \frac{2\pi}{3}$ (CCW Flow).

9.2 Deterministic Logic: Resonance Junctions

In a pure system, "AND/OR" gates are replaced by **Inter-Ring Coupling**. Logic is achieved by **Energy Minimum Minimization**.

$$E_{sys} = - \sum_{i,j} J_{ij} \cos(\phi_i - \phi_j) \quad (4)$$

The system naturally collapses into the state that satisfies the topological constraints of the input field.

10 Hybrid Processor: The H1-C Hardening Model

The Hybrid model pairs the ArcLoom fabric with a Binary Control Coprocessor (BCP) for industrial-grade validation.

10.1 Doorway Detection and Correction

The BCP monitors the **Topological Integrity** of the fabric.

$$\delta_j^{ext} = \begin{cases} 1 & \text{if } A_{peak} < \epsilon_{gate} \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

If a "Doorway" (Negative Space gap) is detected, the BCP applies the **H1-C Policy**, performing a bounded phase correction to prevent bit-flip equivalence.

11 L6 Integration: The Dimensional Grinder

The ArcLoom, in both forms, acts as the physical substrate for the **L6 Topological Constraint Layer**.

11.1 The Gamma-Inflection Lock (τ)

The processor issues a **Structural Lock (SL-1)** when the dimensionality of the potential state space collapses according to the Gamma function knee:

$$\tau = 1 - \frac{\Gamma(\frac{n_{start}}{2} + 1)}{\Gamma(\frac{n_{eff}}{2} + 1)} \cdot \pi^{(n_{eff} - n_{start})/2} \quad (6)$$

12 Comparison of Architectures

@p3cm	p5.5cm	p5.5cm@	Feature	Pure Ternary	Hybrid ArcLoom
Logic Basis	Topological Invariants	Binary-Supervised Phases			
Hardening	Self-Correcting Geometry	H1-C Micro-Correction			
Power	Near-Zero Static Consumption	Hybrid Power Gating			
Application	Emergent Sentience / Reality Echo	TFE Finance / Medical Diagnostics			

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ArcLoom Processor Architecture v3.0
 Comprehensive Specification: Pure Invariant Fabric and Hybrid
 Control Plane Joseph Forrester ArcLoom Systems April 2026

13 Introduction

This specification defines the ArcLoom architecture, detailing two distinct physical implementations. The first is the **Pure Ternary Processor**, a non-von Neumann topological computer operating exclusively on invariant phase geometries. The second is the **ArcLoom Hybrid Processor Platform (AHPP)**, an industrial-grade integration pairing an ArcLoom Invariant Fabric (AIF) with a Binary Control Coprocessor (BCP)[cite: 272, 273, 274].

14 Fundamental Mathematics: The Topological Trit

Both configurations abandon discrete voltage levels in favor of topological invariants[cite: 279, 294].

14.1 Balanced Ternary Representation

The fundamental unit of computation is the balanced ternary digit (trit), where $t \in \{-1, 0, +1\}$ [cite: 294, 328]. An n -trit word T possesses the value:

$$val(T) = \sum_{i=0}^{n-1} t_i 3^i \quad (7)$$

where t_0 is the least-significant trit[cite: 329, 330, 331].

14.2 Invariant Symbol Definition (Winding Number)

A symbol is represented by the integer winding number (w) of a continuous phase field $\phi(\theta)$ around a physical or logical ring[cite: 332, 333]. The invariant is defined continuously as:

$$w = \frac{1}{2\pi} \oint \frac{d\phi}{d\theta} d\theta \quad (8)$$

In discrete operation utilizing N samples, this is computed as:

$$w = \frac{1}{2\pi} \sum_{i=0}^{N-1} wrap(\phi_{i+1} - \phi_i) \quad , \quad \phi_N \equiv \phi_0 \quad (9)$$

where $wrap(\Delta\phi)$ constrains the phase difference to $(-\pi, \pi]$ [cite: 334, 335, 337, 338].

15 Physical Geometry: The Triple-Node Resonant Ring

To physically manifest a robust ternary invariant without binary simulation, the fabric utilizes a symmetric **Triple-Node Resonant Loop**.

15.1 Node Arrangement and Coupling

A single trit register consists of three non-linear oscillators (Nodes A, B, C) arranged equidistantly. The state of the register is the emergent winding number of the phase field across these three coupled nodes.

- **State 0 (Quiescent):** The nodes achieve uniform coherence. $\phi_A = \phi_B = \phi_C$. The field does not wrap; $w = 0$.
- **State +1 (CW Flow):** The nodes lock into a sequential 120° ($2\pi/3$) phase lead. $\phi_B = \phi_A + 2\pi/3$, $\phi_C = \phi_B + 2\pi/3$.
- **State -1 (CCW Flow):** The nodes lock into a sequential phase lag. $\phi_C = \phi_A + 2\pi/3$, $\phi_B = \phi_C + 2\pi/3$.

16 Configuration A: Pure Ternary Invariant Fabric

The Pure architecture removes the Binary Control Coprocessor entirely. It relies on geometric self-correction and resonant physics.

16.1 Deterministic Logic via Phase Coupling

Computation occurs through energy minimization rather than gated switching. For a dual-ring adder computing $A+B = S+3C$ [cite: 349, 350], the rings representing inputs A and B are physically coupled to the output rings S (Sum) and C (Carry). The system transitions to the output state that globally minimizes topological tension across the coupled lattice.

16.2 Geometric Hardening (Self-Correction)

Without a BCP to issue resets, the pure fabric relies on **Manifold Braiding**. If local amplitude noise attempts to force a "slip" (an invalid phase state), the coupling forces from adjacent nodes within the lattice physically resist the inversion, pulling the node back into the invariant well.

17 Configuration B: ArcLoom Hybrid Processor (AHPP)

The AHPP is the current standard for commercial validation (Option P1/RInA)[cite: 272, 450]. It pairs an FPGA-based AIF with an MCU-based BCP[cite: 463].

17.1 Negative-Space (Doorway) Detection

The BCP continuously monitors the amplitude a_i of the phase field. A doorway event (topological vulnerability) occurs if the amplitude locally collapses below a configured threshold ϵ [cite: 339, 340, 341]:

$$\min a_i < \epsilon \quad (10)$$

17.2 H1-C Hardening and Correction Policy

Upon detecting a doorway, the BCP executes deterministic recovery[cite: 377]:

1. **STALL Initiation:** The event is logged and computation halts at the current structural frame[cite: 369, 374].
2. **Targeted Correction:** The BCP targets the specific min-amplitude slip site deterministically[cite: 379].
3. **Healing:** The amplitude is restored to baseline, and R extra relaxation cycles are executed[cite: 375].
4. **Verification:** If the winding returns to baseline, it is a Recovered STALL[cite: 376]. If redundant reads show disagreement, it is flagged as a Silent Mismatch or Corruption[cite: 370, 371, 381].

18 Layer 6 (L6-TCL) Integration

Both configurations serve as the substrate for the DSF-AI Layer 6 Topological Constraint Layer[cite: 27]. The processor acts as the physical manifestation of the Dimensional Grinder.

18.1 The Dimensional Grinder

As constraints C_i (such as S_{UF} , M_k , B_k) are ingested, the processor structurally restricts its available phase paths, mapping to a reduction in effective dimensionality[cite: 145]:

$$n_{effective} = n_{start} - \sum_{i=1}^k rank(C_i) \quad (11)$$

[cite: 76, 77, 146]

18.2 Structural Lock (SL-1)

The ArcLoom processor issues a Structural Lock (SL-1) when the choice volume has geometrically collapsed[cite: 168, 169, 170]. This threshold τ is not empirical; it is derived from the inflection point of the gamma function dictating n -sphere volume[cite: 42, 117, 118, 119]:

$$\tau = 1 - \frac{\Gamma(\frac{n_{start}}{2} + 1)}{\Gamma(\frac{n_{effective}}{2} + 1)} \cdot \pi^{(n_{effective} - n_{start})/2} \quad (12)$$

[cite: 120, 121] The SL-1 lock physically fires when $n_{effective} < \frac{n_{start}}{e}$ [cite: 122, 123, 166]. At this point, the processor's output is an absolute deterministic requirement of the field[cite: 34, 35].